EXP No : 1

DATE - 4-2-2026

VERIFICATION OF KIRCHOFF'S LAWS -

AIM :

to verify kirchoff's voltage law and Kirchoff's current law both theoretically & practically for a given DC circuit

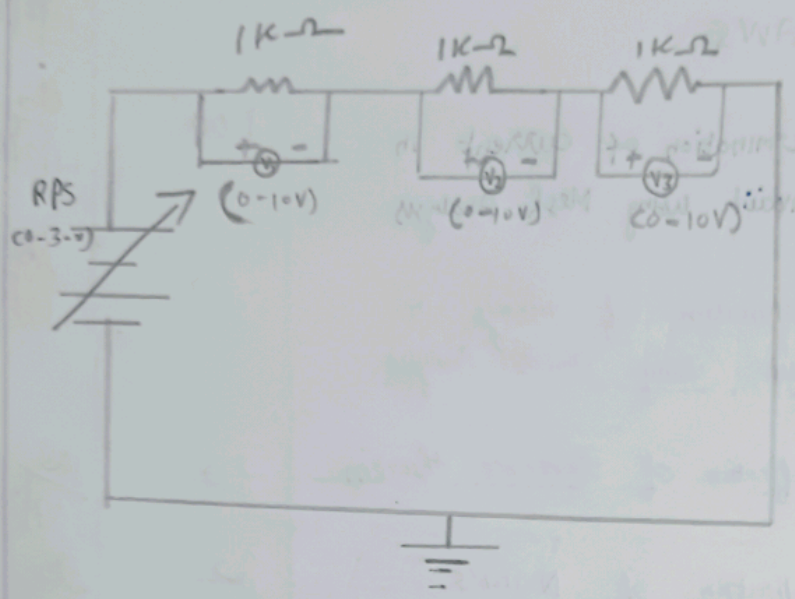
Apparatus Required :

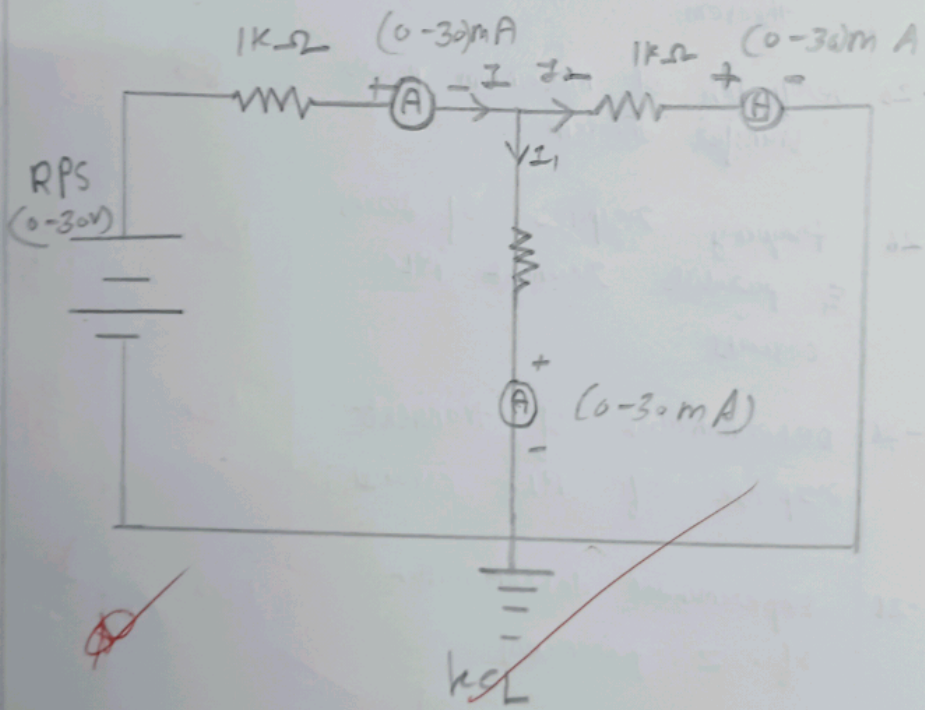
S. No	Apparatus	Specification	quantity
1.	Regulated power supply (RPS)	(0-30V)	1
2.	Voltmeter	(0-30V) MC	3
3.	Ammeter	(0-10mA) MC	3
4.	Resistors	1KΩ	3
5.	Bread board	--	1

Procedure :

1) connect the circuit as per the diagram
2) Switch on the RPS and set the input voltage
3) Note the reading of voltmeters & ammeters
4) vary the input voltage & record all reading
5) Switch off the RPS
6) verify the kirchoff's laws using the recorded values

CIRCUIT DIAGRAM :





TABULAR COLUMN :

V (volts)	V_1 (volts)	V_2 (volts)	V_3 (volts)	$V = V_1 + V_2 + V_3$ (volts)	I (amps)	I_1 (amps)	I_2 (amp)	$I_2 - I_1 + I_2$ (amps)
2	0.7	0.7	0.7	2.1	1.16	0.58	0.58	1.16
6	2.03	2.03	2.03	6.09	3.4	1.7	1.7	3.4
9	3.05	3.05	3.05	9.15	4.8	2.4	2.4	4.8

Approved

11/5/26

CALCULATIONS :

- Practical verification :

  V/I should be constant (ohm's law)

  V should be equal to $(V_1 + V_2 + V_3)$

  I_1 should be equal to $(I_2 + I_3)$

- Theoretical verification :

Kirchoff's voltage law :

i) for V = 2 volts

$V_1 = 0.7$, $V_2 = 0.7$, $V_3 = 0.7$

$V = V_1 + V_2 + V_3 = 0.7 + 0.7 + 0.7 = 2.1V \simeq 2V$

ii) for V = 6 volts

$V_1 = 2.03$, $V_2 = 2.03$, $V_3 = 2.03$

$V = V_1 + V_2 + V_3 = 2.03 + 2.03 + 2.03 = 6.09 \simeq 6V$

iii) For V = 9 volts

$V_1 = 3.05$, $V_2 = 3.05$, $V_3 = 3.05$

$V = V_1 + V_2 + V_3 = 3.05 + 3.05 + 3.05 = 9.15 \simeq 9V$

• Kirchoff's current law :

for $v = 2$, $I = 1.16\,A$, $I_1 = 0.58\,A$, $I_2 = 0.58\,A$

$I = I_1 + I_2 = 0.58 + 0.58 = 1.16\,A$

for $v = 6$, $I = 3.4\,A$, $I_1 = 1.7\,A$, $I_2 = 1.7\,A$

$I = I_1 + I_2 = 1.7 + 1.7 = 3.4$

for $v = 9$, $I = 4.8\,A$, $I_1 = 2.4\,A$, $I_2 = 2.4\,A$

$I = I_1 + I_2 = 2.4 + 2.4 = 4.8\,A$

MARKS

S.No

1.

2.

3.

4.

5.

10

11/2/26

RESULT

current

theoretic

MARKS SPLIT-UP :

S.No	Description	Marks Allocated	Marks obtained
1.	Experimental Setup / connections	20 Marks	20
2.	Observation / Readings	20 Marks	20
3.	Calculations / Results	20 Marks	20
4.	Viva	20 Marks	20
5.	Record	20 Marks	20
		Total	100

100

11/2/26

RESULT :

Thus kirchoff's Voltage law and kirchoff's current law are verified practically & theoretically.